

connected devices rapidly populating the IoT landscape, the need to put an end to security breaches of sensitive and confidential information has never been more critical."

Threats we face

Narang Kishor, principal design architect, Namix Technolabs, explained, "The IoT paradigm is all about connecting everything that exists with everything else. The IoT is a neural network of globally aware, intelligent and heterogeneous devices connected with each other in multiple ways to solve some problems of the society."

"The problem stems when we go overboard with it, like we have, and there is no control mechanism. Earlier, technologies like supervisory control and data acquisition (SCADA) and remote sensing worked in a controlled environment. Now, the applications have opened up to almost every product or service. Applications like smart home and smart healthcare are on-board the IoT. Larger concepts like Smart Cities further add to the complexity. With public Wi-Fi, anybody can connect with anything in the locality because the infrastructure is automated and intelligent.

"Excessive penetration of such technologies also brings about many potential risks. For example, every functioning device is electricity dependent. The moment there is a power failure, life comes to a standstill. On the Smart City network, if the security isn't strong enough, a hacker can access your critical centres and bring the entire nation to a standstill. Today, we don't need a nuclear bomb to destroy a nation, we just need a few good hackers to bring down its power and communication and the nation would be on its knees. A few months ago, Russian hackers used Chinese webcams to attack American servers."

It is estimated that 50 billion devices will be connected by 2020



This proliferation poses new privacy and security risks that must be assessed



Fig. 1: IoT opportunities and challenges (Courtesy: <https://insights.samsung.com/>)

Divyang Hemnani, engineer-embedded systems, Infinium Solutionz, added, "Hackers steal data by infecting the software of IoT devices. They use this infection to modify firmware in the devices. Consequently, all the data can be linked with the hacker's server. Due to the depth of software hacks, people are generally unable to recognise the security vulnerability of their devices."

Sharing his views on the vulnerabilities of IoT systems, Parag Sharma, CEO, Mantra Labs, said, "Security keys are at times used to authenticate the code being executed on an embedded device, or to authenticate the connection from an IoT device to the server. In the event of a security key being compromised, a hacker may be able to intrude into the device and run an unintended malicious code. This is entirely unwanted, and exposes users to serious consequences. Unintended behaviour of a device, and mounting hack attempts from these devices are the threats being faced. While it may not be disastrous for some IoT devices for entertainment to malfunction, a critical life-saving device or a device of strategic importance going haywire can have disproportionate losses associated."

Major layers of IoT architecture

An IoT system consists of a complex multi-tiered architecture, with connectivity and security at the fulcrum

of the setup. It includes the following significant layers:

Edge device and components layer

This layer consists of components like sensors, actuators and controllers embedded in or connected to the devices present in the field or at the edge. In this layer, the data is actually generated, evaluated and digitised.

Network or transport layer

Data digitised in the edge layer is transferred to the central cloud server through gateways. Connected application devices can access the data from the cloud. The cloud provides a Platform as a Service (PaaS), where the data is stored and analysed.

End application layer. This tier constitutes the operational application on the user device (smartphone, computer, smartwatch, etc.). It displays and represents data captured by the sensors and shares over the cloud.

Standards to ensure IoT security

Dr Rishi Mohan Bhatnagar, president, Aeris India, and chairperson of The Institution of Engineering and Technology - IoT panel for India, said, "When it comes to IoT security, three aspects need attention—sensors, network and applications. With billions of devices coming online, we now have that many more vulnerabilities created, even if we ignore the